

changes that are being incorporated into the new version. The mechanical and electronic design of the device has evolved towards a lightweight sensing device that features a revised user interface to integrate an antenna, radar electronics, digital processing and increased battery life.

Ranging buried cable intrusion detection sensor.

This is a covert outdoor perimeter intrusion detection sensor. Buried sensor cables generate an invisible detection field that the system monitors to detect and locate intrusions. Being invisible, it offers the lowest vulnerability to the defeats of any outdoor perimeter intrusion detection sensor while ensuring site aesthetics.

The system uses ported coaxial sensor cables to create an invisible electromagnetic detection field. Equipment within the cables make it possible to transmit energy from the transmitting cable for detection by the corresponding parallel receiving cable. The exact intrusion location (including those occurring simultaneously) can be determined using a coded pulse signal algorithm.

Detection and location of hidden or buried casualties is a difficult task. The need to detect casualties in situations such as collapsed buildings, dense undergrowth and thick smoke is becoming more common and, consequently, there is a need for specialised equipment to deal with associated problems. Devices that register vibrations or sounds that would otherwise go undetected have been found to be the most effective for the location of buried casualties.

Better models of this type of equipment have filtering systems to reduce background noise, making it easier to detect sounds made by trapped casualties. Features and benefits of such a device are:

- Covertly detects and locates perimeter intrusions at a distance of up to 800m per processor
- Pinpoints intrusions up to one metre
- Completely covert; site aesthet-



The electronics inside FINDER

ics remain unchanged

- Operations not affected by vegetation (including grass, shrubs and trees)
- Insensitive to wind, rain, snow, hail, fog, sandstorms, seismic vibrations, acoustics, magnetic effects and blowing debris
- Flexible, software-based zoning
- Communication path redundancy ensures continued perimeter protection in case of a cable cut
- Power and data over sensor cables simplify infrastructure requirements

Drowning detection. Only a limited amount of diving activities can be conducted to search for drowned bodies. A more efficient way to conduct this kind of body recovery activity is to detect the resting frequencies of human body parts—a human body (of height about 180cm) has a frequency of around 3Hz from the side, and from the top (head 25cm) the frequency is around 6Hz. It is necessary to incorporate state-of-the-art underwater acoustics techniques, such as scanning sonar, into operation procedures.

The purpose of the present research is to investigate the application of scanning sonar for underwater human body detection and recovery. Technically, a microphone that can detect very deep frequencies and a ping for everybody (multiple ping frequencies for multiple body parts) is needed. A human body can reflect all dis-

tinctive frequencies because each body part has its own frequency, depending on its position. So, it is important to calculate a frequency profile of all body parts to express the characteristic polynomial.

A sonar surveillance system can detect drowning humans but only in small indoor places (like swimming pools) and not in open water. If a body is lying on the seabed, a very high frequency (> 800kHz) sector scan sonar will work, but with very short ranges (10 to 50 metres, typically), other types of sonar should give a clear image of the legs, arms and body.

Real cases where this technology was incorporated for rescue work has illustrated that the movement of a diver, several suspected targets and morphological features around the targets can be recognised by the scanning sonar. In addition, dimensions, shapes, amount and geographic locations of the targets can be identified for further consideration.

It has been proved indirectly that a scanning sonar can detect drowned human bodies successfully. The detection and identification ability of the scanning sonar can be dramatically improved if optical apparatus such as remotely-operated vehicles (ROVs) or towed-operated vehicles (TOVs) are incorporated into operation procedures.

Conclusion

Electronic devices currently being used to search for drowning people or those buried under debris work with microphones and cameras. These can only help locate entrapped people who are capable of making sounds or are visible beneath ruins. In the future, scientists want to complement these resources with advanced electronic-cum-chemical sensors to detect buried and drowning human beings so that they can be rescued alive. In this regard, drones and robots equipped with electronic-cum-gas sensors can also be used, allowing difficult-to-reach or inaccessible areas also to be searched. **EFY**